Santiago Barrera-Thirlwall

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Education

McMaster University **September 2022 – April 2027**
Bachelor of Engineering, Mechanical Engineering (Co-op) Hamilton, ON

- Placed 1st of 150+ students in the robot competition for MECHENG 2C04, a robotics design course.

Work Experience

BBA Consultants **January 2026 – August 2026**
Mechanical Engineering Co-Op Toronto, ON

- Coordinated a cyclone cluster modification from concept to IFC, including calculations, site visits, and SOW.
- Modeled a new distributor and piping for a carbon safety screen replacement, verifying the distributor with FEA.
- Ran FEA in Ansys Mechanical to resolve fatigue cracking in a leach tank agitator support structure.
- Wrote the scope of work for a metal removal system at a Nevada gold mine and answered bidder questions.
- Maintained 96% billable utilization, averaging 166 billable hours per month after onboarding.

Teck Resources – Highland Valley Copper Mine **May 2025 – December 2025**
Mechanical Engineering Co-Op Logan Lake, BC

- Redesigned AG mill discharge chutes to extend liner life, developing a custom cast liner package with a foundry.
- Engineered a roof access walkway, ship ladder, and platform from 3D scan data, replacing a long exterior ladder.
- Designed lifting devices and stands for gearboxes, HVAC units, and submersible pumps.
- Built a parametric SolidWorks tool to automate design of the site's standard handrails.

Enedym Inc. **September 2024 – April 2025**
Part-Time Mechanical Designer Hamilton, ON

- Retained after co-op to design housings, busbars, and other components for a new generation of inverters.
- Assembled, tested, troubleshot, and repaired switched reluctance motors, inverters, and converted EVs.

Enedym Inc. **May 2024 – August 2024**
Mechanical Engineering Co-Op Hamilton, ON

- Redesigned aluminum motor housings from machined to extruded for production, cutting cost and material waste.
- Redesigned end caps to match, rerouting cooling channels and designing seals per the Parker O-Ring Handbook.
- Designed mounts and covers, installed, and test-drove motors and inverters on diesel-to-EV converted airport tugs.

McMaster Manufacturing Research Institute **May 2023 – August 2023**
Research Assistant II Co-Op Hamilton, ON

- Programmed and operated industrial CNC equipment using Fusion 360 CAM and hand-written G-code.

Columbia International College **September 2022 – Present**
Activities Instructor & FIRST Robotics Mentor (Part-Time) Hamilton, ON

- Mentor students through the design, fabrication, and testing of 150 lb FIRST Robotics competition robots.
- Run the school's CNC routers and train students to operate and maintain them.

Extracurricular Experience

McMaster Formula Electric **September 2022 – Present**
Drivetrain Member; Drivetrain Lead; Manufacturing Member Hamilton, ON

- Promoted to Drivetrain Lead 7 months after joining; ran design and testing.
- Led design of a custom in-hub planetary gearbox, 29% shorter than the prior design and fixing its oil leaks.
- Recruited members, assigned deliverables, and reviewed all gear calculations, CAD, and drawings.
- Helped build four carbon fiber monocoques by vacuum infusion, plus a chassis mold from machined foam positives.

Skills

CAD/CAE: SolidWorks, Ansys Mechanical, Fusion 360, AutoCAD, Inventor, Onshape, KISSsoft/KISSsys, PDM
Design & Manufacturing: GD&T, DFM, CNC machining, composites, 3D printing
Certifications: CPR & First Aid, Fall Protection, Rigging, SCBA, Confined Spaces, Lockout/Tagout